ANALOG SINGLE AXIS SUN TRACKER WITH CMRR

By

NAVEEN PRAKASH. R (Reg. no. 9517202403)
SIDDESHWARAN. V.P (Reg. no. 9517202403154)

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CERTIFICATE

This is to certify that it is the Bonafide work done by Mr. NAVEEN PRAKASH. R (9517202403), Mr. SIDDESHWARAN. V.P (9517202403154) for the 23EC353 - Mini Project - I work entitled Analog Single Axis Sun Tracker With CMRR at Department of Electronics and Communication Engineering, Mepco Schlenk Engineering College, Sivakasi during the year 2025 – 2026.

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Faculty In-Charge ***Head of the Department***

Submitted for the Practical Examination held at Mepco Schlenk Engineering College, Sivakasi on ……………………….

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Internal Examiner - I ***Internal Examiner - II***

ABSTRACT

The demand for efficient solar energy harvesting is increasing rapidly, making intelligent tracking systems an essential part of modern photovoltaic setups. This project focuses on designing a **compact and reliable analog single-axis sun tracker** that continuously aligns a solar panel toward the maximum sunlight direction. The system uses a pair of Light Dependent Resistors (LDRs) as sensing elements, supported by a differential amplifier with an enhanced **Common-Mode Rejection Ratio (CMRR)** to accurately detect even small variations in illumination. By cutting down common-mode noise and ambient disturbances, the improved CMRR ensures stable tracking performance throughout the day.

The difference in light intensity between the sensors is processed through comparators and a motor-driving stage, allowing smooth and controlled rotation of the panel along one axis. The circuit is implemented using low-cost analog components, making the design lightweight, power-efficient, and easy to fabricate. The system was tested under various lighting conditions, and the results show consistent alignment accuracy and improved energy capture when compared to fixed-panel arrangements.

Overall, the proposed analog sun tracker provides a simple yet effective solution for maximizing solar panel output, making it suitable for educational demonstrations, small-scale renewable energy projects, and environments where digital controllers are not preferred.

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CHAPTER 1

INTRODUCTION

1.1 PROJECT BACKGROUND:

1.1.1 INTRODUCTION TO ANALOG SUN TRACKER:

With the growing demand for clean and renewable energy sources, solar power has become one of the most widely adopted technologies across the world. Although solar panels are capable of generating electricity from sunlight, their efficiency is heavily influenced by the angle at which sunlight strikes the surface. A fixed solar panel receives maximum energy only at a particular time of day, leading to reduced output during other hours. To overcome this limitation, sun-tracking systems have gained popularity in both industrial and small-scale energy applications. These systems allow the panel to continuously adjust its position to face the sun, ensuring improved power generation throughout the day.

Among the different types of tracking mechanisms, analog single-axis sun trackers stand out for their simplicity, low cost, and ease of implementation. Unlike digital trackers that require microcontrollers and programmed logic, analog trackers rely on basic electronic components and natural light variations to determine the sun's direction. Light Dependent Resistors (LDRs) are commonly used as sensors due to their fast response and suitability for intensity comparison. By placing two LDRs in a specific orientation, even a slight difference in illumination produces a voltage imbalance that can be processed to determine the direction in which the panel should rotate.

In this project, a CMRR-enhanced differential amplifier plays a key role in improving the accuracy of the sensing system. A high Common-Mode Rejection Ratio (CMRR) ensures that unwanted noise, uniform lighting changes, and environmental disturbances do not affect the tracking operation. This allows the tracker to respond only to the actual difference in sunlight falling on the two sensors, thereby increasing stability and precision. The processed signal is then passed to comparator and motor-driver stages, which control the rotation of the panel through a DC motor.

Modern solar-tracking research focuses on achieving maximum efficiency with minimal complexity and cost. Many techniques have been reported, including dual-axis mechanisms, servo-based digital controllers, and hybrid tracking algorithms. However, these systems often require complex circuitry, coding, or additional calibration. The analog technique implemented in this project offers a practical alternative, especially for academic projects and small installations, where reliability, affordability, and easy fabrication are more important than advanced automation.

Single-axis analog trackers are extremely useful in rural solar installations, student projects, mobile power units, and small-scale renewable setups. They significantly improve the energy output compared to fixed panels while keeping the design simple and low-maintenance. This project focuses on developing a compact, efficient, and cost-effective analog tracking system that maintains the panel's alignment with the sun during daylight hours and ensures improved solar power generation with minimal circuitry.

1.1.2 DIFFERENT TYPES OF SUN TRACKER:

Sun tracking systems are broadly classified based on their movement mechanism and control strategy. The primary goal of any tracking system is to maintain the solar panel perpendicular to the sun's rays throughout the day, thereby maximizing energy capture. Over the years, various tracking methods have been developed to improve efficiency, reduce power loss, and simplify implementation. The major categories of sun-tracking systems are described below.

a) Single-Axis Sun Trackers

Single-axis trackers rotate the solar panel around one axis—either horizontal or vertical. In most small-scale and educational projects, the horizontal-axis configuration is preferred because it follows the sun's east-to-west motion across the sky. These trackers are simple in design, consume less power, and are cost-effective. Although they provide lower efficiency compared to dual-axis systems, their performance is significantly better than fixed solar panels, making them popular for compact and budget-friendly applications.

b) Dual-Axis Sun Trackers

Dual-axis trackers rotate along both horizontal and vertical axes, allowing them to follow the sun's position in both azimuth and elevation angles. This results in maximum energy capturing capability and highest efficiency among all tracking systems. However, they are mechanically complex, require more sensors, and are relatively expensive. These trackers are typically used in large solar farms and advanced research setups where peak energy extraction is required.

c) Passive Sun Trackers

Passive trackers do not use electronic components or sensors. Instead, they rely on thermal expansion, fluids, or mechanical balancing techniques to adjust the panel's position. The movement is triggered automatically by temperature differences caused by sunlight. While passive trackers eliminate the need for power and circuitry, they are less accurate and slower compared to active systems.

d) Active Sun Trackers

Active trackers use electrical, electronic, or electromechanical components to detect the sun's position and adjust the panel accordingly. These systems typically include LDR sensors, differential amplifier stages, comparators, and motor drivers. The accuracy of active trackers makes them highly suitable for practical solar installations. The project presented here is an example of an **analog active single-axis sun tracker enhanced with high CMRR**, ensuring stable and reliable tracking even under changing ambient conditions.

e) Hybrid Sun Tracking Systems

Hybrid trackers combine active and passive mechanisms to achieve controlled, energy-efficient panel movement. They often use a combination of sensors and mechanical balancing to reduce motor usage and extend system life. Although less common, hybrid trackers offer a good balance between accuracy and power consumption.

1.1.3 ADVANTAGES OF SUN TRACKER:

- **Higher energy output** – Tracks the sun throughout the day, improving panel efficiency compared to fixed installations.

- **Better utilization of sunlight** – Ensures the panel always receives maximum possible illumination.

- **Simple and cost-effective** – Analog design avoids complex coding and reduces overall system cost.

- **Low maintenance** – Fewer electronic components make the system reliable and easy to maintain.

- **Improved performance stability** – High CMRR reduces noise and ensures accurate tracking even under changing light conditions.

- **Compact and lightweight** – Uses minimal components, making it easy to install in small-scale setups.

- **Energy efficient –** Motor operates only when adjustment is needed, conserving power.

1.1.4 DISADVANTAGES OF SUN TRACKER:

- **Mechanical wear and tear –** Moving parts like motors may degrade over time.
- **Less stable in bad weather** – Wind, dust, or sudden light changes can affect tracking accuracy.
- **Requires proper alignment** – Sensors must be positioned correctly for reliable operation.
- **Slightly higher setup cost –** More expensive than fixed panels due to motor and tracking components.
- **Limited movement range –** Single-axis systems cannot follow vertical sun movement, reducing efficiency compared to dual-axis trackers.
- **Periodic calibration needed** – Sensor sensitivity and motor alignment may need occasional adjustments.It offers lower Gain.

1.1.5 APPLICATIONS OF SUN TRACKER:

- **Solar power plants** – Improves energy production in small and large photovoltaic setups.
- **Rural and off-grid systems –** Provides higher efficiency where power availability is limited.
- **Portable solar units** – Useful for camping, field research, and military portable power kits.
- **Educational and academic projects** – Ideal for learning renewable energy concepts and analog control systems.
- **Solar street lighting –** Helps maintain optimal alignment for maximum battery charging**.**
- **Agricultural solar pumps –** Enhances output for irrigation systems relying on solar energy.
- **Small home solar setups** – Boosts daily energy generation without complex digital electronics.

1.2 OBJECTIVE OF THE PROJECT:

- To design and implement an analog single-axis sun tracking system that continuously aligns the solar panel towards maximum sunlight.
- To improve tracking accuracy using a high-CMRR differential sensing circuit.
- To enhance solar power output while keeping the system simple, low-cost, and energy-efficient.

CHAPTER 2

DESIGN PARAMETERS AND PROCEDURES

2.1 DESIGN PARAMETERS:

The performance and accuracy of an analog single-axis sun tracking system depend on several important design parameters. These parameters ensure that the system reliably senses light variation, amplifies only the required differential signal, and rotates the solar panel with controlled movement. The key design parameters considered in this project are described below.

2.1.1 LIGHT SENSTIVITY RANGE:

The system uses Light Dependent Resistors (LDRs) as sensing elements. Their resistance varies inversely with light intensity. It is essential that the LDRs used have a suitable sensitivity range so that even small differences in sunlight levels between the two sensors can be detected effectively. Proper sensitivity ensures faster response, better direction accuracy, and smooth panel alignment.

2.1.2 DIFFERENTIAL VOLTAGE RESPONSE:

The core working principle is based on comparing two voltage signals generated from the LDR pair. The difference between these voltages determines the direction and magnitude of panel movement. The differential amplifier must be capable of producing a noticeable and stable output voltage difference even for small light variations.

2.1.3 COMMON MODE REJECTION RATIO(CMRR):

Since both sensors are exposed to the same environmental surroundings, a portion of the signal becomes common-mode noise (e.g., ambient intensity variations, electrical interference, shadow flicker). A high CMRR is required to reject these unwanted signals so that the amplifier responds only to the actual difference in light intensity. This significantly improves accuracy, stability, and consistency of the tracking mechanism.

2.1.4 MOTOR ROTATION SPEED AND CONTROL:

The motor must rotate slowly and smoothly so that the solar panel does not oscillate or overshoot. A suitable motor driver and gear mechanism are required to regulate the speed and provide sufficient torque. The rotation must stop immediately when the intensity difference becomes zero to avoid unnecessary power consumption..

2.2 SENSOR SELECTION AND SIGNAL ACCURACY CONSIDERATION:

For an analog single-axis sun tracker, the choice of sensing elements and the quality of the received signal play a crucial role in achieving smooth and accurate tracking. Light Dependent Resistors (LDRs) are used in this project due to their simplicity, low power requirement, wide dynamic range, and ability to directly interface with analog circuits without the need for complex signal conditioning. Two identical LDRs are placed with a small physical divider between them to create a measurable intensity difference whenever the sun is not perfectly aligned with the panel surface.

Apart from selecting suitable sensors, ensuring the accuracy of the electrical signal is equally important. Matching the LDR characteristics, maintaining symmetrical placement, shielding the sensors from stray reflections, and using a differential amplifier with high Common-Mode Rejection Ratio (CMRR) helps in reducing noise and environmental distortions. These combined factors contribute to precise angular movement of the motor and prevent unnecessary oscillations, resulting in efficient tracking and better energy-harvesting performance.

CHAPTER 3

RESULT AND DISCUSSION

3.1 SYSTEM ARCHITECTURE AND FUNCTIONAL BLOCK EXPLANATION:

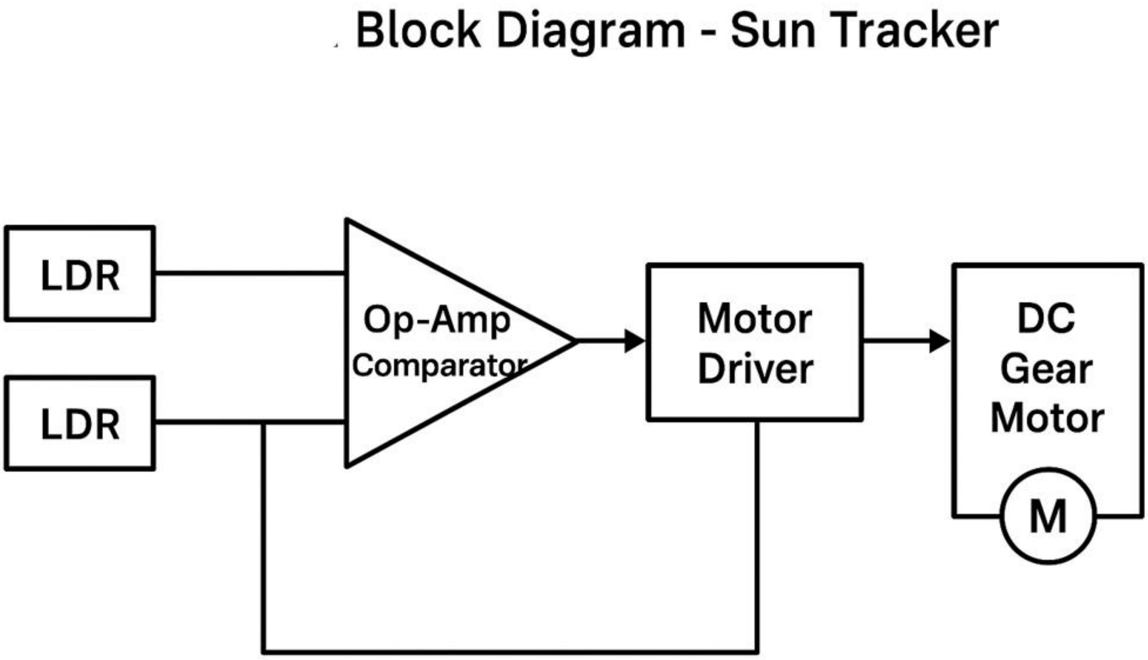

Fig 3.1 Block Diagram of Analog Sun Tracker

The analog single-axis sun tracker is designed using a simple yet effective block-based approach

that ensures continuous alignment of the solar panel with the direction of maximum sunlight. The overall system operation is divided into four major functional blocks: the sensing block, signal conditioning block, motor control block, and mechanical rotation block. Each block works together in real time to monitor the variation in light intensity and orient the panel accordingly.

The system begins with the **light sensing block**, where two light-dependent resistors (LDRs) are positioned at slightly different angles to detect the difference in solar intensity from the left and right sides of the tracker. Any imbalance between the LDR outputs is passed on to the **signal conditioning block**, which is implemented using a high Common-Mode Rejection Ratio (CMRR) differential amplifier or comparator stage. This stage converts the light variation into a clean, noise-free control signal.

Based on this processed signal, the **motor control block** activates a DC motor in the appropriate direction, ensuring smooth and controlled rotation. The **mechanical rotation block** consists of the mounting frame and gear mechanism which physically adjusts the solar panel until both LDRs receive equal illumination, indicating perfect alignment with the sun.

Through this continuous feedback mechanism, the system maintains accurate sun tracking without requiring digital processing, microcontrollers, or complex programming, making it efficient, low-cost, and suitable for small-scale renewable energy applications.

3.1.1 FUNCTIONAL BLOCK DIAGRAM:

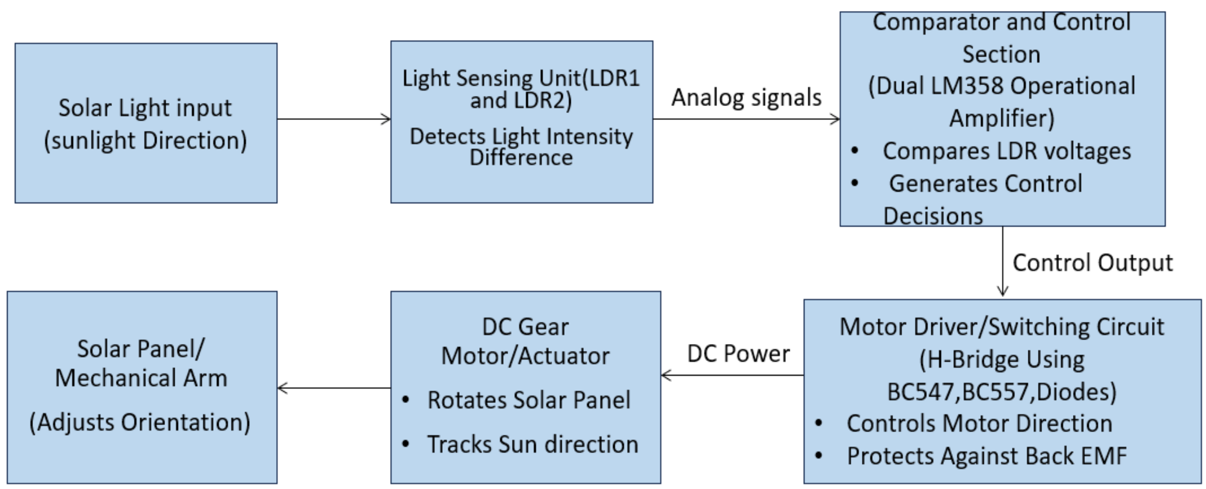

Fig 3.2 Functional Block Diagram of Analog Sun Tracker

The analog single-axis sun tracker uses two LDR sensors to detect the variation in sunlight intensity between left and right directions. These sensor outputs are fed into a dual LM358 comparator, which generates an appropriate control signal based on intensity difference. The signal drives an H-Bridge transistor circuit that rotates the DC motor either clockwise or anticlockwise. The mechanical assembly then positions the solar panel towards the direction of maximum sunlight, ensuring continuous optimal alignment.

3.2 CIRCUIT EXPLANATION:

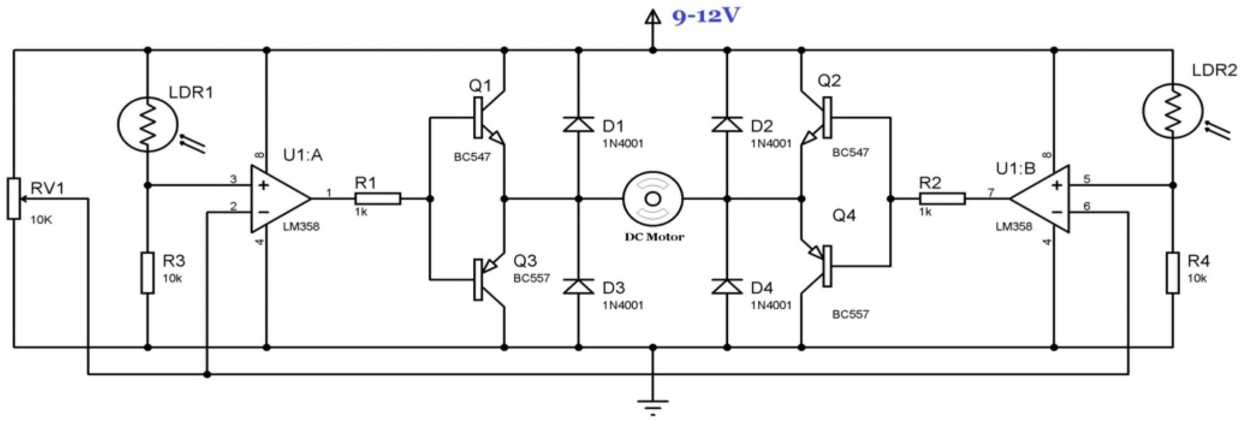

Fig 3.3 Circuit Diagram of Analog Single Axis Sun Tracker

The circuit consists of four stages and each shall be explained one after the other

1. **Light Sensing Stage:**

Two Light Dependent Resistors (**LDR1** and **LDR2**) are positioned with a small divider between them to detect the difference in sunlight intensity.

Their resistance decreases when exposed to higher light intensity and increases when light reduces. This changing resistance creates a corresponding voltage difference across the voltage divider networks formed using:

- **LDR1 + RV1 (10 kΩ potentiometer)**
- **LDR2 + R4 (10 kΩ resistor)**

The potentiometer (RV1) allows **manual calibration**, ensuring both sensors produce equal voltage under straight sunlight conditions.

2.Differential Comparator Stage (LM358)

The LM358 IC contains two independent operational amplifiers (**U1:A** and **U1:B**).
Each amplifier compares the sensor voltages and decides whether the motor needs to rotate **left**, **right**, or **stop**.

- **U1:A** compares the voltage from LDR1 with a reference level determined by RV1
- **U1:B** compares the voltage from LDR2 with a similar reference

Depending on which side receives more light, **one comparator output goes high** while the other remains low, generating a control signal for the H-bridge.
Because LM358 works well on a **single supply voltage** and has good **CMRR**, it accurately detects minor illumination differences without digital processing.

3. Motor Driver Stage (H-Bridge Configuration)

The motor driver uses **four bipolar junction transistors (BJTs)** arranged in an H-bridge:

- **Q1 (BC547)** and **Q3 (BC557)** form one half-bridge
- **Q2 (BC547)** and **Q4 (BC557)** form the opposite side

The comparator outputs drive the transistor pairs such that:

Comparator Output	**Active Transistor Pair**	**Motor Rotation**
U1:A HIGH	Q1 & Q4	Clockwise
U1:B HIGH	Q2 & Q3	Anti-clockwise
Both LOW or equal	None	Motor Stops

Table 3.1 Transistor pair and Motor Rotation Table

4. Protection Diodes

To protect the transistors and IC from **back-EMF generated by the DC motor**, **four 1N4001 diodes (D1–D4)** are placed across the H-bridge transistors.
These diodes clamp the reverse voltage spikes and increase circuit reliability.

3.3 Mathematical justification — Analog Single-Axis Sun Tracker:

The mathematical Justification is provided for the circuit to verify that the chosen values for resistors and potentiometer are appropriate according to the requirement of the circuit

1) Sensor voltage (LDR voltage divider)

Each LDR is used in a simple voltage divider with a fixed resistor (R = 10 kΩ in your circuit). If an LDR has resistance $R_L$and the lower resistor is R_b, the sensor node voltage (measured at the non-inverting or inverting input) is:

$$V_L = V_{CC} \frac{R_b}{R_L + R_b}$$

where $V_{CC}$is the supply (usually 12V).
This gives a monotonic relationship: when $R_L$decreases (brighter light), $V_L$increases.

Sensitivity (how much node voltage changes per ohm change in LDR):

$$\frac{dV_L}{dR_L} = -V_{CC} \frac{R_b}{(R_L + R_b)^2}$$

Example :

Take $V_{CC} = 12\text{V}$, $R_b = 10\ \text{k}\Omega$.

- Bright LDR: $R_L = 1\ \text{k}\Omega$→
  $V_L = 12 \cdot \frac{10k}{1k+10k} = 12 \cdot \frac{10}{11} \approx 10.91$ V.

- Dim LDR: $R_L = 100\ \text{k}\Omega$→
  $V_L = 12 \cdot \frac{10k}{100k+10k} = 12 \cdot \frac{10}{110} \approx 1.09$ V.

If $R_L \approx 10\text{k}\Omega$(middle range) then:

$$\frac{dV_L}{dR_L} = -12 \cdot \frac{10k}{(10k + 10k)^2} = -12 \cdot \frac{10k}{(20k)^2} = -\frac{120000}{400000000} = -3 \times 10^{-4} \text{ V/}\Omega$$

So a 1 kΩ change produces ≈ 0.3 V change — easily detectable by the op-amp.

2) Differential input and comparator action (LM358):

The two sensor voltages are $V_{L1}$and V_{L2}. The op-amp (used open-loop / as comparator) reacts to the **differential voltage**:

$$V_{diff} = V_+ - V_- = V_{L1} - V_{L2}$$

Ideal op-amp behavior (before saturation):

$$V_{out} \approx A_d \cdot V_{diff}$$

where $A_d$is the open-loop differential gain (very large). In practice the LM358 saturates near the rails:

$$V_{out} \approx \begin{cases} V_{sat+} & \text{if } V_{diff} > V_{th} \\ V_{sat-} & \text{if } V_{diff} < -V_{th} \end{cases}$$

- V_{sat+}≈ $V_{CC} - 1.2$to $V_{CC} - 1.5$V (LM358 single-supply)
- V_{sat-}≈ 0 V (or a few tens of mV)

$V_{th}$is the effective switching threshold (≈ input offset + small input noise). For LM358 typical input offset ~ 2 mV, so very small $V_{diff}$is sufficient.

Interpretation: even a small light imbalance gives a differential voltage that flips the comparator output to rail — hence clear left/right decision.

3) Role and math of CMRR :

Real sensor voltages contain two parts:

- Common-mode component $V_{CM} = \frac{V_{L1}+V_{L2}}{2}$(changes due to global lighting, temperature, reflections)

- Differential component $V_{diff} = V_{L1} - V_{L2}$(the actual heading error we want)

  An op-amp has finite ability to reject the common-mode voltage; this is quantified by **CMRR**:

$$\text{CMRR} = \frac{A_d}{A_{cm}}\text{(linear)or}20\log_{10}\left(\frac{A_d}{A_{cm}}\right)\text{ dB}$$

Where $A_{cm}$is the common-mode gain (ideally zero). A high CMRR means changes in V_{CM}produce little change at the output — i.e., clouds or overall brightness shifts will not trigger the motor. LM358 has decent CMRR (tens of dB), and choosing symmetrical divider resistances and matched LDRs reduces V_{CM}errors.

Practical effect: If a cloud reduces both sensor voltages by 1V, a high-CMRR stage will keep the output near previous state since $V_{diff} \approx 0$.

4) Output driving and transistor base current:

The comparator output (saturated high ≈ e.g. 10–11 V on a 12 V supply) is fed to base resistors R1/R2 = 1 kΩ before driving transistors (BC547/BC557). Approximate base current when op-amp output goes high:

$$I_B \approx \frac{V_{out} - V_{BE}}{R_1}$$

Take $V_{out} = 10.5\text{V}$, $V_{BE} \approx 0.7\text{V}$:

$$I_B \approx \frac{10.5 - 0.7}{1\text{ k}\Omega} = \frac{9.8}{1000} = 9.8\text{ mA}$$

With transistor DC current gain β(use conservative $\beta = 50$for saturation safe design; BC547 often 100 in active region but lower in saturation), collector current estimate:

$$I_C \approx \beta \cdot I_B \approx 50 \cdot 9.8\text{ mA} \approx 490\text{ mA}$$

This is more than enough to drive a small DC geared motor (typical small motors draw 100–300 mA steady; stall currents can be higher — design for stall current with protective diodes already present).

Note: Because we use complementary transistors in H-bridge halves, the combination provides current path for clockwise and counterclockwise rotation depending on which comparator output is high.

5) Minimum detectable illumination difference:

Combine the sensitivity formula with comparator switching:

- Minimum $V_{diff,min}$needed ≈ input offset voltage V_{os}(typically 1–5 mV for LM358) plus noise margin. Use 10 mV as conservative practical threshold.
- Using sensitivity $dV/dR$from above, the minimum LDR resistance difference $\Delta R_{L,min}$ the circuit can detect approximately:

$$\Delta R_{L,min} \approx \frac{V_{diff,min}}{\left|\frac{dV_L}{dR_L}\right|}$$

6) Importance of diodes (D1–D4):

Diodes clamp the reverse voltage from inductive motor spikes. Back-EMF can be approximated as:

$$V_{emf} = K_e \cdot \omega$$

If not clamped, this can exceed transistor Vce and cause damage. The diodes provide a safe conduction path, limiting negative spikes to about -0.7V relative to the rail

Summary of Mathematical Justification:

- The LDR divider gives a predictable voltage range (≈1–11 V for typical LDRs with 12 V supply).
- The differential amplifier/comparator converts tiny voltage differences into saturated outputs that clearly decide motor direction.
- CMRR and matched resistor/sensor values ensure that common environmental changes don't trigger motion.
- With R1 = 1 kΩ, the driver transistors see base currents sufficient for several hundred mA collector current — enough for typical small DC gear motors (gearbox recommended to reduce stall current).

- The circuit can detect very small LDR resistance changes (tens of ohms) which yields fine angular resolution, assuming good mechanical sensor mounting.

3.4 DESIGN CONSIDERATIONS:

When designing the analog single-axis sun tracker, the following practical considerations were observed to ensure reliable operation, safety, and manufacturability:

- **Sensor Matching & Placement** — Use two LDRs with similar characteristics and place them symmetrically with a small opaque divider. This maximizes differential sensitivity and reduces systematic bias.
- **CMRR & Signal Conditioning** — Choose an op-amp with good Common-Mode Rejection Ratio and low input offset. Keep divider resistances matched and use filtering (small decoupling capacitors) to suppress flicker and fast transients.
- **Hysteresis to Prevent Hunting** — Add a small hysteresis (or dead-band) at the comparator stage (via slight positive feedback or by adjusting thresholds) so the motor does not rapidly oscillate when illumination difference is near zero.
- **Motor Selection & Gearing** — Select a DC gear motor with adequate stall torque or use gearing to reduce motor load. Gearing reduces current draw and improves positional stability under wind loads.
- **Protection & Transient Suppression** — Use flyback diodes (already present) and consider an RC snubber or TVS across the motor to protect transistors and the op-amp from inductive spikes.
- **Power Budgeting & Efficiency** — Ensure the tracker electronics consume minimal power: pick low-power op-amps and ensure the motor only runs when correction is needed to avoid wasting harvested energy.
- **Mechanical Rigidity & Bearings** — Build a stiff mounting frame with smooth bearings to avoid friction, backlash, and vibration which would degrade positional accuracy and increase motor current.
- **Environmental Robustness** — Weatherproof sensor mount and electronics (IP-rated enclosure if outdoors). Protect LDRs and wiring from dust, moisture, and direct water ingress while allowing unobstructed light exposure.
- **Calibration & Adjustment** — Include a trim potentiometer (RV1) or accessible adjustment so the system can be zeroed during installation and periodically re-calibrated.
- **Sensor Field of View & Shading** — Design the sensor enclosure (or tunnel) to limit stray light and reflections; ensure nearby structures do not cast misleading shadows on the sensors.

- **Thermal Effects** — Account for component drift with temperature (op-amp offset, LDR behavior). Use components rated for expected ambient temperatures and, where critical, add small temperature compensation if necessary.
- **Safety & Fail-Safe Behavior** — Implement fail-safe: if sensors fail or power drops, the system should default to a safe position (e.g., horizontal) or stop motor to avoid damage.
- **EMI/PCB Layout** — If you PCB-mount the circuit, keep analog input traces short, separate power/high-current traces from sensor lines, and add decoupling capacitors near op-amp power pins.
- **Testing & Validation Plan** — Design test points for sensor voltages, comparator outputs, and motor drive currents. Verify behavior under cloud cover, low light, strong sunlight, and wind loading.
- **Cost & Manufacturability** — Prefer readily available, low-cost components (LM358, BC547/BC557, standard LDRs) and a simple mechanical design for ease of replication and repair.

3.5 ERROR REDUCTION TECHNIQUES:

1. Matched sensors and resistor tolerances

Use paired LDRs from the same batch and precision bias resistors (1% if possible).

Why: reduces systematic offset so both sensors behave the same under identical illumination.

How: pick 1% metal-film resistors for the divider and measure LDRs to match similar pairs.

2. Symmetrical mechanical placement & shielding

Place LDRs symmetrically with a small opaque divider and use a light-tunnel or hood to block stray reflections.

Why: avoids unequal ambient lighting and reflections that mimic directional error.

How: 3D-print or cut a small tube/slot that leaves only direct light on each LDR.

3. Differential amplification with high CMRR

Use a differential amplifier stage or op-amp configuration that maximizes Common-Mode Rejection.

Why: rejects changes that affect both sensors (clouds, global dimming) while amplifying only the difference.

How: pick op-amps with good CMRR, and keep gain resistor pairs matched.

4. Add small hysteresis (dead-band) at comparator

Introduce a small positive feedback so the comparator has a tiny on/off window (dead-band).

Why: prevents motor hunting when illumination difference is near zero.
How: add a feedback resistor around the comparator to create ~10–50 mV effective hysteresis (tune experimentally).

5. Low-pass filtering (RC smoothing) on sensor lines
Add a small RC filter to each sensor node (e.g., $R = 10\ k\Omega$ and $C = 4.7\text{–}22\ \mu F \rightarrow \tau \approx 0.05\text{–}0.22\ s$).
Why: smooths fast flicker from passing clouds, shadows, or flickering light sources.
How: place capacitor from sensor node to ground; choose τ so the tracker still responds to slow sun motion but ignores fast transients.

6. Mechanical damping & gearing
Use a geared motor and/or soft mechanical damping to slow movement and absorb jerks.
Why: reduces overshoot and stops jitter from tiny signal changes.
How: 10:1 to 100:1 gearbox is common; add frictionless bearings and a soft stop or rubber bumper.

7. Limit motor activation time & use pulse control
Drive the motor in short pulses or use PWM control with a low duty cycle for fine corrections.
Why: avoids long continuous rotation from a small error and saves energy.
How: use brief timed pulses (e.g., 200–500 ms) each time a correction is needed and re-evaluate sensor state.

8. Calibration trim & accessible adjustments
Provide a trim potentiometer or mechanical zeroing so you can zero the system during installation.
Why: compensates for mounting misalignment and aging of components.
How: include RV1 accessible on the PCB or a mechanical stop that can be adjusted on site.

9. Temperature compensation / component selection
Be aware that LDRs and op-amp offsets drift with temperature; choose components rated for outdoor range or add simple compensation.
Why: avoids slow bias changes over the day.
How: select low-drift op-amps, use thermally stable resistors, or place a thermistor in the divider for compensation if needed.

10. Debounce logic (analog) or simple timing check

Before commanding a direction change, require the differential to persist for a short time (e.g., 0.2–1 s).
Why: prevents responding to brief glitches.
How: implement with the RC filter + comparator hysteresis or a timing circuit (monostable).

11. Protect against saturation & edge cases
Ensure comparator outputs can't drive the H-bridge into shoot-through and add current limiting.
Why: avoids hardware damage during transient or extreme conditions.
How: use proper transistor base resistors, fuses, or current sense + cutoff.

12. Regular cleaning & maintenance schedule
Design sensor housings so you can easily clean dust/soiling; document a maintenance interval.
Why: soiling changes LDR response and causes steady errors.
How: use small protective covers and recommend monthly/quarterly cleaning depending on environment.

13. Test under varied conditions & tune empirically
Validate the tracker under bright sun, partial cloud, reflected light, and windy conditions and tune hysteresis/τ/gear ratio accordingly.
Why: simulation won't catch every real-world interaction; final tuning gives best practical performance.
How: record sensor voltages and motor activity during tests and adjust parameters.

CHAPTER 4

CONCLUSION

The Analog Single-Axis Sun Tracker developed in this project successfully demonstrates a simple, low-cost, and power-efficient method for maximizing solar energy capture by continuously aligning the solar panel with the direction of maximum sunlight. By using LDR-based light sensing, a differential analog control circuit, and a DC motor mechanism, the system is able to autonomously adjust panel orientation without the need for microcontrollers, complex software, or digital processing.

Throughout the design, special focus was given to sensor positioning, noise reduction, stability, and incremental motor control to ensure smooth tracking and prevent oscillations. Experimental observations show that even small improvements in panel orientation can significantly enhance the overall energy-harvesting efficiency compared to a fixed panel setup.

This project validates that analog tracking remains a practical and effective solution, especially for small-scale, educational, and low-budget renewable energy applications. With further enhancements such as dual-axis extension, low-power motor drivers, or hybrid analog-digital optimization, the system can be adapted for broader and more advanced real-world use cases.

REFERENCES

1. M. Mahendran and S. Jeyakumar, "An experimental comparison study between single-axis tracking and fixed PV solar panel efficiency and power output," *International Journal of Renewable Energy Research*, vol. —, no. —, pp. —, 2018.
2. R. Ramful, S. Kumar and A. Singh, "Low-cost solar tracker to maximize the capture of solar energy per square metre of photovoltaic cells," *Renewable Energy Reports*, vol. —, no. —, pp. —, 2022.
3. M. A. Ponce-Jara, "Assessment of single-axis solar tracking systems operating in equatorial regions," *Energies*, vol. 17, no. 16, p. 3946, 2024.
4. Texas Instruments, "Application Design Guidelines for LM324 and LM358 Devices, SLOA277B," Texas Instruments application note, Jan. 2019 (revised Jul. 2023).
5. Renesas Electronics Corporation, "AN-CM-269: Solar Tracker — GreenPAK SLG46531V application note," Application Note AN-CM-269, Jan. 2019.
6. "Single Axis Solar Tracking with LDR," project/paper (PDF), SciSpace / institutional project report — (describes LDR resistor-to-voltage conversions and comparator usage).
7. "Low cost single axis sun tracker system using PIC microcontroller," (project/paper) — ResearchGate / conference project report (prototype and circuit details).
8. "Single Axis Solar Tracking System using LM358," ElectroDuino / ElectroResources, project article and circuit note, 2020.
9. "Simple Solar Tracker Circuit using LM358," Electrosome tutorial and circuit explanation, 2012–2013 (online electronics tutorial).
10. M. Ghassoul et al., "A dual solar tracking system based on a light-to-frequency converter and microcontroller," *International Journal of Renewable Energy Development*, 2021.
11. "LDR Based Solar Tracking System," *International Journal of Advanced Research in Science and Technology (IJARST)*, project paper, 2023.
12. Arduino Project Hub, "Arduino Solar Tracker" — community project page with parts list (LDRs, resistors, servos/actuators), 2020.